

The Gradient Maximizes the Directional Derivative

David Puelz

The University of Austin

1 Setup

Directional derivative. For a differentiable function $f(x, y)$ and a **unit** vector $\mathbf{u} = (u_1, u_2)$ (so $u_1^2 + u_2^2 = 1$), the directional derivative of f in the direction $\mathbf{u}$ is

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u} = \frac{\partial f}{\partial x} u_1 + \frac{\partial f}{\partial y} u_2.$$

This gives the rate of change of f as we move in direction $\mathbf{u}$.

Question. Among all unit vectors $\mathbf{u}$, which direction gives the *largest* directional derivative? We will show, using first-order conditions and ordinary derivatives, that the answer is: the direction of the gradient ∇f .

2 Two Dimensions: First-Order Conditions with the Constraint

Step 1. We maximize $h(u_1, u_2) = \frac{\partial f}{\partial x} u_1 + \frac{\partial f}{\partial y} u_2$ subject to the constraint $u_1^2 + u_2^2 = 1$. Use the constraint to write $u_2 = \sqrt{1 - u_1^2}$ (we consider the upper half of the unit circle; the lower half gives the minimum). Then

$$h(u_1) = \frac{\partial f}{\partial x} u_1 + \frac{\partial f}{\partial y} \sqrt{1 - u_1^2}.$$

Step 2. First-order condition: set $\frac{dh}{du_1} = 0$. Using the chain rule on $\sqrt{1 - u_1^2}$ (derivative of outer $\sqrt{\cdot}$ times derivative of inner $1 - u_1^2$):

$$\frac{dh}{du_1} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} \cdot \frac{1}{2}(1 - u_1^2)^{-1/2} \cdot (-2u_1) = \frac{\partial f}{\partial x} - \frac{\partial f}{\partial y} \frac{u_1}{\sqrt{1 - u_1^2}} = 0.$$

Step 3. Since $\sqrt{1 - u_1^2} = u_2$, we have

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} \frac{u_1}{u_2} \Rightarrow \frac{\partial f}{\partial x} u_2 = \frac{\partial f}{\partial y} u_1.$$

Step 4. Why does this imply $\mathbf{u} \propto \nabla f$? Two vectors (u_1, u_2) and (a, b) are proportional if there exists $\lambda \neq 0$ such that $u_1 = \lambda a$ and $u_2 = \lambda b$.

Equivalently, the ratios of components match: $u_1/u_2 = a/b$ (when $u_2, b \neq 0$).

From $\frac{\partial f}{\partial x} u_2 = \frac{\partial f}{\partial y} u_1$, we can write (assuming $\frac{\partial f}{\partial y} \neq 0$):

$$\frac{u_1}{\partial f / \partial x} = \frac{u_2}{\partial f / \partial y} = \lambda$$

for some constant λ . Hence $u_1 = \lambda \frac{\partial f}{\partial x}$ and $u_2 = \lambda \frac{\partial f}{\partial y}$, so

$$(u_1, u_2) = \lambda \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right) = \lambda \nabla f.$$

The constraint $u_1^2 + u_2^2 = 1$ then gives $\lambda^2 |\nabla f|^2 = 1$, so $|\lambda| = 1/|\nabla f|$. There are two critical points: $\lambda = 1/|\nabla f|$ (where $\mathbf{u}$ points with ∇f) and $\lambda = -1/|\nabla f|$ (where $\mathbf{u}$ points against ∇f). The former gives $D_{\mathbf{u}}f = |\nabla f|$ (maximum); the latter gives $D_{\mathbf{u}}f = -|\nabla f|$ (minimum). We want the maximum, so we take $\lambda = 1/|\nabla f|$ and obtain

$$\mathbf{u}^* = \frac{\nabla f}{|\nabla f|}.$$

Step 5. The maximum value of the directional derivative is

$$D_{\mathbf{u}^*}f = \nabla f \cdot \frac{\nabla f}{|\nabla f|} = \frac{|\nabla f|^2}{|\nabla f|} = |\nabla f|.$$

Second derivative check. Differentiate $\frac{dh}{du_1}$ again. Using $u_2 = \sqrt{1 - u_1^2}$,

we have $\frac{du_2}{du_1} = -\frac{u_1}{u_2}$ (chain rule). By the quotient rule:

$$\frac{d}{du_1} \left(-\frac{u_1}{u_2} \right) = \frac{-u_2 + u_1 (du_2/du_1)}{u_2^2} = \frac{-u_2 - u_1^2/u_2}{u_2^2} = -\frac{u_2^2 + u_1^2}{u_2^3} = -\frac{1}{u_2^3},$$

since $u_1^2 + u_2^2 = 1$. So $h''(u_1) = \frac{\partial f}{\partial y} \cdot \left(-\frac{1}{u_2^3} \right)$. At the critical point,

$u_2 = \frac{\partial f / \partial y}{|\nabla f|}$ has the same sign as $\frac{\partial f}{\partial y}$, so $\frac{\partial f}{\partial y} / u_2^3 > 0$ and $h'' < 0$.

Hence we have a local maximum.

3 Summary

Proposition 4. Among all unit vectors $\mathbf{u}$, the directional derivative $D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ is maximized when $\mathbf{u}$ points in the direction of ∇f , i.e., $\mathbf{u} = \nabla f / |\nabla f|$. The maximum value is $|\nabla f|$.

Interpretation. The gradient points in the direction of steepest ascent. Moving in that direction increases f at the fastest possible rate (per unit distance).